

MIZORAM UNIVERSITY

Department of Life Sciences

Syllabus for Ph.D. Course



**Pachhunga University College
Mizoram University
Aizawl 796001
Mizoram**

MIZORAM UNIVERSITY

Ph.D. Course Work in Life Sciences

SUBJECTS OF STUDY AND SCHEME OF EXAMINATION

Part	Title of the Paper	Paper code	Credit	Exam duration (h)	Internal test	Seminar/assignment	End Semester Marks	Total
I	Research Methodology	LSC-601	4	3	20	20	60	100
II	Instrumentation: Tools and Techniques	LSC-602	4	3	20	20	60	100
III	Research and Publication Ethics (RPE)	LSC-603	2	3	20	20	60	100
IV	Taxonomy	LSC-604 (A)	4	3	20	20	60	100
	Parasitology and Microbiology	LSC-604 (B)						
	Biochemistry and Phytochemistry	LSC-604 (C)						
	TOTAL		14					400

N.B.

1. Part I to III (601, 602 and 603) are common to all students.
2. Part IV has three optional papers out of which a student shall opt only one.

Evaluation

- Total credit a student must earn is 14 and the total final examination marks is 400.
- The course will be completed in one semester.
- There will be two internal assessments, as:
 - 1) One test which will carry 20 marks
 - 2) One assignment in the form of report or seminar
- End semester examination will carry 60 marks, which may include dissertation as a component.
- Pass mark in each paper is 50% of the full marks.
- The final marks will be converted in 10-point scale as:

7.00–10.00	=	O (Outstanding)
6.00–6.99	=	A (Pass)
5.00–5.99	=	B (Pass)
- A student must secure at least B in all the papers.

LSC-601
Research Methodology

L:T:P = 3:2:0
Credits = 5
Marks = 100

Unit I

Literature review; Finding research material - Science Direct, Annual Reviews, Springer, impact factor and citation index; Identification and scope of research problems; design of experiments; data management; Scientific writings -synopsis, thesis and research paper; IPR and bio-piracy.

Unit II

Organization of laboratory; Safety measures - use and handling of hazardous chemicals, disposal of biological and radioactive wastes.

Unit III

Statistical methods: measurements of central tendency, standard deviation, standard error, student's t-Test, Chi square test, null hypothesis, correlation, regression, ANOVA, use of statistical software (SPSS, STATISTICA).

Unit IV

Introduction to search engine/Data Base: NCBI, EMBL, DDBJ, SwissProt, TIGR; Multiple sequence alignment, Phylogenetic analysis, Primer designing, Protein structure analysis.

Suggested readings:

- Bailey, N.T.J. *Statistical Methods in Biology* (3rd edition). Cambridge University Press, New York, 1997.
- Parsons, C.J. *Thesis and Project Work. A guide to Research and Writing*. George Allen and Unwin Ltd., London, 1973.
- Rajagopalan, K., *Essentials of Computer Fundamentals*, CBS Publishers and Distributors, New Delhi, 1991.
- Snedecor, G.W., William, G. *Statistical Methods*, Harvard University, Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi, 1975.
- Mount, D. W. *Bioinformatics: Sequence and Genome Analysis* (2nd edition). Cold Spring Harbor Laboratory Press, New York, USA, 2001.
- Basevanis A.D., Quellette, B.F.F. *Bioinformatics: A Practical Guide to the Analysis of Genes and Proteins*. John Wiley, Malden, 2005.
- Shantharam,S, Montgomery, J.F. *Biotechnology, Biosafety and Biodiversity. Scientific and Ethical Issues for Sustainable Development*. Oxford and IBH Publishing Co.,New Delhi, 1999.

LSC-602
Instrumentation: Tools and Techniques

L:T:P = 3:2:0
Credits = 5
Marks = 100

Unit I

Biochemical calculations; Preparation of solutions; Culture media preparation and sterilization; Electrochemical techniques; Remote sensing and its applications.

Unit II

Centrifugation, Spectrophotometry- UV-VIS and atomic absorption; Chromatography – TLC, HPLC and GC; Microscopy- Light and electron microscopes.

Unit III

DNA and protein isolation; Agarose and Polyacrylamide gel electrophoresis; Restriction mapping; PCR - RAPD, RFLP, AFLP; RT-PCR and gel documentation; Blotting techniques.

Unit IV

Enzyme assay- specific activity and kinetics; Enzyme-linked immunosorbent assay; Application of radioisotopes in biology.

Suggested readings:

- Talwar, G.P. *Handbook of Practical Immunology*. Vikas Publishing House Pvt. Ltd., New Delhi, 1983.
- Sabins, F.F. *Remote Sensing Principles and Interpretation*, II Ed., W.H., Freeman and Company, New York, 1987.
- Segel, I.H. *Biochemical Calculations*, (2nd edition), John Wiley and Sons, Asia Pvt Ltd., 2004
- Wilson, K., Walker, J. *Practical Biochemistry: Principle and Techniques*, Cambridge University Press, NY, 2005.
- Cappuccino, J.G., Sherman, N. *Microbiology : A Laboratory Manual*. Dorling Kindersley, Delhi, 2002.
- Sambrook J., Russell D.W. *Molecular Cloning : A Laboratory Manual. Vol I, II and III*. Cold Spring Harbor Laboratory, New York. 2001.

LSC-603
Research and Publication Ethics (RPE)

L:T:P = 1:0:1
Credits = 2
Marks = 100

THEORY

RPE 01: PHILOSOPHY AND ETHICS (3 hrs.)

1. Introduction to philosophy: definition, nature and scope, concept, branches
2. Ethics: definition, moral philosophy, nature of moral judgements and reactions

RPE 02: SCIENTIFIC CONDUCT (5 hrs.)

1. Ethics with respect to science and research
2. Intellectual honesty and research integrity
3. Scientific misconducts: Falsification, Fabrication, and Plagiarism (FFP)
4. Redundant publications: duplicate and overlapping publications, salami slicing
5. Selective reporting and misrepresentation of data

RPE 03: PUBLICATION ETHICS (7 hrs.)

1. Publication ethics: definition, introduction and importance
2. Best practices/standards setting initiatives and guidelines: COPE, WAME, etc.
3. Conflicts of interest
4. Publication misconduct: definition, concept, problems leading to unethical conduct and vice versa, types
5. Violation of publication ethics, authorship and contributorship
6. Identification of publication misconduct, complaints and appeals
7. Predatory publishers and journals

PRACTICE

RPE 04: OPEN-ACCESS PUBLISHING (4 hrs.)

1. Open access publications and initiatives
2. SHERPA/RoMEO online resource to check publisher copyright & self-archiving policies
3. Software tool to identify predatory publications developed by SPPU
4. Journal finder/tools viz. JANE, Elsevier Journal Finder, Springer Journal Suggester, etc.

RPE 05: PUBLICATION MISCONDUCT (4 hrs.)

- A. Group Discussions (2 hrs.)
 1. Subject specific ethical issues, FFP, authorship
 2. Conflicts of interest
 3. Complaints and appeals: examples and fraud from India and abroad
- B. Software tools (2 hrs.)
 1. Use of plagiarism software like Turnitin, Urkund and other open source software tools

RPE 06: DATABASES AND RESEARCH METRICS (7 hrs.)

- A. Databases (4 hrs.)
 1. Indexing databases
 2. Citation databases: Web of Science, Scopus, etc.
- B. Research Metrics (3 hrs.)
 1. Impact Factor of journal as per Journal Citation Report, SNIP, SJR, IPP, Cite Score
 2. Metrics: h-index, g index, i10 index, altmetrics

Suggested readings:

Bird, A. (2006). *Philosophy of Science*. Routledge.

MacIntyre, Alasdair (1967). *A Short History of Ethics*. London.

Chaddah, P. (2018). *Ethics in Competitive Research: Do not get scooped; do not get plagiarized*, ISBN:978-9387480865

National Academy of Sciences, National Academy of Engineering and Institute of Medicine. (2009). *On a Scientist: A Guide to Responsible Conduct in Research*: Third Edition. National Academies Press.

Resnik, D.B. (2011). *What is ethics in research & why is it important*. National Institute of Environmental Health Sciences, 1-10. <https://www.niehs.nih.gov/research/resources/bioethics/whatis/index.cfm>

Beall, J. (2012). Predatory publishers are corrupting open access. *Nature*, 489(7415), 179-179. <https://doi.org/10.1038/489179a>

Indian National Science Academy (INSA), *Ethics in Science Education, Research and Governance* (2019), ISBN:978-81-939482-I-7. [http://www.insaindia.res.in/pdf/Ethics Book.pdf](http://www.insaindia.res.in/pdf/Ethics%20Book.pdf)

LSC-604(A)
Taxonomy

L:T:P = 3:2:0
Credits = 5
Marks = 100

Unit I

Taxonomic procedures: Taxonomic collections, preservation, curation, process of identification. Typification. Taxonomic keys, merits and demerits. Evaluation of biodiversity indices. - Evaluation of Shannon - Weiner Index. - Evaluation of Dominance Index. Similarity and Dissimilarity Index.

Unit II

Classification system of Plants, Animals, Protozoa, Bacteria and Viruses. Role of IUCN, IUCN, ICN, ICTV. Formation of Scientific names of various Taxa.

Unit III

Molecular techniques in Taxonomy- DNA-DNA hybridization, DNA fingerprinting, RFLP analysis, PCR sequencing, Realtime PCR. DNA, RNA, Protein sequence editing, Genome sequencing and editing, multiple sequence alignment, Phylogenetic analysis and common tree-building methods.

Unit IV

Genome and proteome databases: Gen Bank, PDB and Pubmed; NCBI genome database; BLAST(N), BLAST(P); BOLD system. Primer design and submission of sequence in database.

Suggested readings:

- Pellens, R., Grandcolas, P. (2016). *Biodiversity Conservation and Phylogenetic Systematics*. Springer.
- Watson, M. F., Lyal, C. H. L., Pendry, C. A. (2015). *Descriptive Taxonomy: The Foundation of Biodiversity Research*. Cambridge University Press.
- Mount, D. W. (2001). *Bioinformatics: Sequence and Genome Analysis* (2nd edition). Cold Spring Harbor Laboratory Press, New York, USA.
- Kapoor V.C (2008). *Theory and Practice of Animal Taxonomy* (6th Edition). Oxford and IBH Publishing.
- Hillis, D.M., Moritz, C. & Mable, B. K (1996). *Molecular Systematics* (2nd edition). Sinauer Associates, Inc. Publishers, Sunderland, Massachusetts U.S.A.
- Balfour, A. (2016). *Principles of Plant and Animal Taxonomy*. Syrawood Publishing House.

LSC-604(B)
Parasitology and Microbiology

L:T:P = 3:2:0
Credits = 5
Marks = 100

Unit I

History of microbiology. Staining Methods – Simple, Gram, Acid-fast, Spore, Granular, Capsular, Flagellar and Fat bodies; Microbial diversity and extremophiles; Structure, transmission, and pathology of medically and agriculturally important virus, bacteria and fungi; Viroids.

Unit II

Medically important Protozoans, arthropods and other helminthic diseases, diagnostic methods; Acanthocephalans; Vector borne diseases; entomopathogenic parasites; Host-Pathogen Interactions and Immune Responses to Infection.

Unit III

Identification techniques for various pathogens, reliability and applications. Microbial sources of pharmaceutically important compounds; Principles of Chemical Signalling and Communication by Microbes, Quorum sensing.

Unit IV

Bacterial Recombination; Sensitivity and Antibiotic resistance; Microbes as a Biological Control agent; Bioremediation, biotransformation and biodegradation. Cancer and mutation- types and diagnostic methods.

Suggested readings:

- Beltz LA (2014). *Emerging Infectious Diseases: A Guide to Diseases, Causative Agents, and surveillance*. Wiley Publication.
- Bennett, J. E., Dolin, R., Blazer, M. R. (2014). *Mandell, Douglas and Bennett's Principles and Practice of Infectious Diseases Vol 1 & 2* (8th edition). Elsevier Health.
- Chatterjee KD (2011). *Parasitology, Protozoology and Helminthology*. Alkem (13th Edition)
- Holt, J S., Kreig N R., Sneath P. H. A and Williams ST. (1994). *Bergey's Manual of Determinative Bacteriology* (9th edition) Williams and Wilkins, Baltimore.
- Leventhal, R (2012). *Medical Parasitology*. F.A. Davis Company, Pennsylvania, United States (3rd Edition)
- Madigan, M., Martinko, J., Stahl D. & David Clark D. (1997) *Brock Biology of Microorganisms* (Thirteenth Edition). Pearson International edition.
- Pelczar Jr, M.J. Chan, E.C.S and Krei N.R (1993) *Microbiology*. McGraw Hill, New York.
- Prescott LM, Harley JP and Klein DA (2003) *Microbiology* (10th edition). McGraw Hill, New York.

LSC-604(C)
Biochemistry and Phytochemistry

L:T:P = 3:2:0
Credits = 5
Marks = 100

Unit I

Amino acids: classification and structure; Protein synthesis and DNA replication, transcription, translation; Lipids: general structure; Structure and classification of carbohydrate (mono-, oligo- and polysaccharides).

Unit II

Physiological role of plant growth hormones; Photoperiodism; Vernalization; Seed dormancy; Senescence; Nitrogen metabolism; Biological nitrogen fixation, assimilation of nitrate and ammonia; Enzymes: classification, nomenclature, mechanism of action.

Unit III

Secondary plant metabolites and bioactive compounds; Collection of samples and drying; selection of plant and plant parts for phytochemical analysis.

Unit IV

Methods of extraction, retrieval of solvent; Qualitative and quantitative estimation of phytochemicals; Applications of chromatography and spectroscopy.

Suggested readings:

- Buchanan B, Gruissem W and Jones R.L. (Eds.) (2015). *Biochemistry and Molecular Biology of Plants*. 2nd Edition, Paperback, Wiley-Blackwell.
- Cone, E.E., Stumpf, P.K. and Bruening, G. (2006). *Outline of Biochemistry* 4th Ed. John Wiley and Sons Inc.
- Harborne J.B. (1973). *Phytochemical Methods*. John Wiley & Sons, New York.
- Harborne J.B. (1998). *Phytochemical Methods – A guide to modern techniques of plant analysis*. Chapman and Hall Publication, London.
- Sivasankar B. (2012). *Instrumental Methods of Analysis*. Oxford University Press.